

Program Name: Bachelor of Computer Applications -Online

Semester: 1

1. Prerequisite: Basic knowledge of Computer

2. Rationale: The objective of this course is to familiarize students with concepts of fundamentals of information technology and detailed working of computer and its application.

3. Course Content:

Sr.	Topics	Weightage
1	Overview • Introduction • Characteristics computers • Evolution computers Generation of computers • Classification of computers • Application of computers • Computer Architecture • Computer Peripherals	13 %
2	Input devices: • Introduction, • Types of input devices, • Optical character recognition, • Optical Mark Recognition, • Magnetic ink character recognition, • Bar code reader, Output devices: • Introduction, • Classification of output devices, Computer software: • Introduction, • software definition, • relationship between software and hardware, • software categories, Computer languages: • Introduction, • Evolution of programming languages, • classification of programming languages, • Features of a good programming language,	13 %
3	Number System: • Introduction- Decimal, Binary, Octal and Hexadecimal. • Inter-Conversions, Addition, Subtraction, Multiplication and Division in Binary Number System. • 1's and 2's Complement method in Binary Number System. Subtraction using 1's and 2's Compliment • Weighted Number System - Binary Coded Decimal (BCD), Addition of BCD Numbers. • Non-Weighted Number System – Applications, Excess-3, Gray code Conversions - Gray and Binary Codes • Fixed point and Floating point	21 %
4	Computer and Communication Fundamentals of Gates: • Basic gates, Derived gates and Universal gates (Design). Combinational and Sequential logic circuits • Half adder, Full adder, • Half -subtractor and Full-subtractor. Flip-Flops – • SR, D, JK, JK Master Slave, T Flip-flops, • Encoders, decoder Multiplexer- and Demultiplexer 4 to 1 line.	19 %
5	Memory Organization: Primary memory: Introduction, • memory representation, • memory hierarchy, • Randomaccess memory, • Types of RAM, Read-only memory, • Types of ROM. Secondary Storage: • Introduction, • classification, • magnetic tape, • magnetic disk, • Optical disk, • Magneto-optical disk, • Memory stick, Universalserial bus, • Mass storage devices.	15 %
6	Internet basics: • Evolution, Basic internet terms, • getting connected to internet, • internet Applications • Types of Networks • Topologies	6 %
7	Microsoft office - Microsoft office	13 %

4. Reference Books

- Digital Logic and Computer Design**
By Morris Mano | PHI
- Introduction to Information Technology**
By ITL Education Solution Limited | Pearson Education | 2012
- MS OFFICE 2007**
By Vikas Gupta | Wiley

4. **Computer Fundamentals**
By Anita Goel | Pearson Education | 2011
5. **Digital Fundamentals**
By Thomas L Floyd | Pearson

5. Course Outcome:

After the completion of the course, the students will be able to:

1. Discuss various number system.
2. Evaluate and simplify Boolean expression and logic circuit.
3. Understand essential concepts of sequential logic and combinational logic.
4. Describe basics of memory organization.
5. Utilize various features of Microsoft office tools.

6. List of Practical

1. Case study on salary calculation

Calculate Allowance based on given Condition. 1. HRA is 10% on Basic Salary if Salary more than 20000. 2. DA is on 25% on Basic Salary. 3. Medical Allowance (MA) = Executives get MA Rs 1000, Officers get MA Rs 700 & Assistants get MA Rs 500 4. Calculate Gross Salary. Gross Salary = Total of Basic + HRA + DA + MA 5. Calculate Professional Tax Upto 5000 = 0, upto 1000 = 60, upto 15000 = 100 & over 15000 = 150 6. Calculate Annual Salary 7. Calculate Income Tax Upto 100000 = 0, then 50000 = 10%, then 100000 = 20%, over 250000 = 30% Hint: Formula =IF(K2<=100000,0,IF(K2<=150000,(J2-100000)*10%,IF(K2<=250000,(K2-150000)*20%+5000,(K2-250000)*30%+25000))) 8. Calculate Net Salary Payable Annual salary - income tax

2. Formatting alignment and creating table

1. Type in the Title Microsoft Word Computer Training Manual 2. Text formatting: Times New Roman font, size 14, Bold and Blue. Paragraph formatting: Align Center. 3. Type in the first paragraph. Text formatting: Arial font, size 11. Paragraph formatting: Align Justify, First Line Indent at 0.5" 4. Type the notes.

3. Word art and clip art

Prepare visiting card for caterer service in word 2007 Prepare interactive word document. (Apply all formatting style)

4. Macro creating macro

5. Invitation letter format

The format of invitation is as shown below: Anand Institute of Information Science, Shri. Ramkrishna Seva Mandal Opp. Town Hall, Anand 388 001 Ph. No. (02696) 266062 To, The Director/Principal, "The name of Institute" "The address of Institute"

6. Work sheet exercise

1. Insert a column 'Number of Teams' between columns 'Year' and 'Tickets sold' with values '32', '38', '39'. 2. Insert a row between row 3 and row 4 with values '1998', '37', '23,290', '1,200,490'. 3. Delete column 'Revenue'. 4. Rename the Sheet1 with name 'Format cells'. 5. Delete Sheet3. 6. Hide row 4. 7. Insert a sheet and rename it with name 'insdel'.

7. Table exercise

Complete the following tasks: a) Widen the first column to 15. b) Add a row beneath the details on Southampton to show the average monthly rainfall. c) Add a new column after the June rainfall statistics to show the total rainfall in each city over the period. d) The rainfall in Birmingham during March should be 58. e) Insert a new row between the rows holding the London and Sheffield rainfall statistics. Enter the following details: Newcastle 65 63 57 50 39 21 f) Copy the appropriate formula to obtain the total rainfall for Newcastle during the period.

8. Table column exercises

1. Change the column width of column B to 15. 2. Change column width of column D to G to 20. 3. Change column width of column A and B to 14. 4. Calculate Total Sales for each item and store result in column D. Hint: Total sales = Quantity * Unit Price. 5. Calculate Total Sales for all the items and store result in cell B6. 6. Copy Unit Price for PC in cell D7. Move Total sales from cell B6 to D8.

9. Insert remove columns of table

Complete the following tasks: a) Add a Units Used column to show the number of units of electricity used by each customer (Hint: Subtract the Previous Reading from the Present Reading). b) The cost of one unit of electricity is Rs.0.08. Add a Unit Cost column to show the cost of one unit. (This column will contain 0.08 in all of the relevant cells). c) Add a Units Charge column to show the total cost of the units used by each customer. (Hint: Unit Cost * Units Used) d) There is a standing charge of Rs.13.60 on each customer's account. Add a column to display this Standing Charge. (This column will contain Rs.13.60 in all of the relevant cells).

10. Math functions

The functions and commands required to solve the following assignment are as follows: Enter data - labels and values 1. Editing cell contents 2. Saving a spreadsheet 3. Altering column widths 4. Using the SUM function 5. Adding a new row after the last row of data 6. Adding a new column after the last column of data 7. Copying a formula 8. Using the AVERAGE, MIN, MAX function 9. Inserting a new row between existing rows Inserting a new column between existing columns

11. Table formatting using background color

Format the Student Grades so that your spreadsheet looks like the one below (you can use different colours, if you like).

12. Calculate total sale and commission

based on given details in table

13. Filter data of excel sheet

1. Count number of order in Boston. 2. Count number of Microwave order. 3. Count number of journeys with truck 3. 4. Count number of Peter White journeys. 5. How many times is no. of items less than 20. 6. Display sum of refrigerator items. 7. Display sum of washing machine items. 8. Display sum of items transported by truck 4. 9. Sum of items transported by trucks. 10. Number of microvawe orders in Boston . 11. Number of Peter White journeys with truck 1. 12. Number of orders in Boston after 2/3/2013: 13. Number of orders between 2/3/2013 and 2/6/2013: 14. sum of microwaves transported to NY: 15. sum of items transported to Pittsburgh by truck 1: 16. sum of items ordered between 2/3/2013 and 2/6/2013: 17. sum of items transported to NY, Baltimore and Philadelphia:

14. Conditional formatting

do conditional formatting on the excel sheet in given data

15. Sorting

sort given data of excel sheet

16. Typing excercise

aq1 qa sw2ws de3ed fr4rf gt5tg queen 11 queens 1 apple 11 apples 2 wishes 22 wishes 2 swims 22 swims eddies 33 eddies 3 deeds 33 deeds 4 roses 44 roses 4 fish 44 fish tugs 55 tugs 5 goats 55 goats

17. Water mark and header footer

inserting and removing

18. Power point presentation

creating presentation

19. PPT

add timing and sound effects

20. Access create data base, tables

create db, tables

21. Access,relations between tables

relations between tables